



AAPP012-4-2-RWDD RESPONSIVE WEB DESIGN & DEVELOPMENT

GROUP ASSIGNMENT

Intake

UCDF2408ICT/UCDF2408ICT(DI)/UCDF2408ICT(ITR)/UCDF2408ICT(SE)

Date Assigned

Week 2

Date Due

Week 13

Course Learning Outcomes	Assignment Component
CLO1 Demonstrate the principles and technologies of the Internet and World Wide Web. (C3, PLO2)	Documentation
CLO2 Construct an effective and interactive web application by using the scripting languages. (P3, PLO3)	Implementation

1. Assignment Brief

Your Web Development Team is tasked with developing a fully functional **sustainability challenge web application**. Your team must select one sustainability theme (examples listed below) and design an application that allows the APU community to participate in the challenge, log their progress, track achievements, and include **additional functionalities that reflect the unique nature of the chosen theme**. The application must support at least **three user roles**, each with distinct access rights and functionalities.

Suggested Sustainability Themes (choose one, or propose your own for approval):

1. **Reducing Plastic Use** – e.g., tracking reusable bottles, containers, bags.
2. **Saving Energy** – e.g., monitoring electricity/water usage, energy-saving tips.
3. **Carpooling & Sustainable Transport** – e.g., ride-sharing, cycling, public transport usage.
4. **Food Waste Reduction** – e.g., surplus food sharing, meal portion tracking.

5. **Green Points & Rewards** – e.g., earning points/badges for eco-friendly actions.
6. **Recycling & Upcycling** – e.g., waste sorting, DIY upcycling guides.
7. **Campus Green Initiatives** – e.g., tree planting, clean-up campaigns, volunteer logging.
8. **Sustainable Events** – e.g., eco-friendly event planning, tracking waste at events.

Note: You are encouraged to be creative. Your team may either select a theme from the list above or propose a related sustainability challenge for lecturer approval.

2. General Requirements

- i. **Client-side scripting:** The application should effectively utilize client-side scripting languages, specifically HTML5, CSS, and JavaScript.
- ii. **Server-side scripting:** PHP should be used as the server-side scripting language.
- iii. **Database Management:** MySQL should be employed for database management.
- iv. **Interface Design:** The interface design should adhere to good design principles and must be responsive.
- v. **Navigation:** The application should feature a fully functional and user-friendly navigation system.
- vi. **Multimedia Content:** Where necessary, appropriate multimedia content should be incorporated.

3. Deliverables

- i. **Proposal (Due Week 4)**
 - Title of the web app project.
 - List at least three user roles and the web functionalities related to them.
 - Assign group members in charge of each role and functionality.
 - Provide appropriate unique functionalities tailored to your chosen sustainability theme.
- ii. **Preliminary Draft (Due Week 7)**
 - Flowchart – illustrating system processes and interactions.
 - Entity Relationship Diagram (ERD) – showing the database structure.
 - Data Dictionary – describing database tables, fields, and relationships.
 - Wireframe – for the main pages of the web application.
 - Navigation Map (Site Map) – outlining the structure and navigation flow of the website.
- iii. **Final Report (Due Week 13)**

Format:

 - Typeface: Times New Roman (bold, italic, underline permitted for emphasis).
 - Font size: 12 (except for titles and headings).
 - Spacing: 1.5 lines.
 - Alignment: Justified.
 - Headers and footers may be used for additional information.
 - All pages must be numbered.
 - A cover page is required, including: Group Member Name & Student ID, Intake Code, Subject, Project Title.

Final Report Content:

- Table of Contents – with page numbers.
- Introduction and Project Plan – objectives, scope, intended users, Gantt chart.
- Design – final Flowchart, ERD, Data Dictionary, Wireframe, Navigation Map.
- Implementation – steps, at least two code snippets per member, screenshots of major pages.
- Conclusions – review work, validate objectives, discuss limitations & improvements.
- References – APA style, focusing on websites, sustainability, and future internet use.

iv. Web App Project (Due Week 13)

The web app should include the following:

- Appropriate graphics, including some created by your team.
- All files must be in PHP format and use HTML5, responsive to different devices.
- Style sheets may be embedded, inline, or external.
- JavaScript should be used to enhance interactivity and functionality.
- MySQL database should align with your ERD and data dictionary.
- The application must clearly demonstrate the three user roles identified in your proposal.
- Ensure usability and accessibility for the target users.

v. Presentation (Starting from Week 13)

- A 30-minute online demonstration via Microsoft Teams.
- All group members must present and explain their contribution.

vi. Suggested Gantt Chart

Task / Milestone	Week													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Group formation & theme selection	■	■												
Proposal preparation & submission			■	■										
Requirement gathering & analysis					■	■								
Preliminary draft (ERD, flowchart, wireframe)						■	■	■						
Implementation – Database & PHP setup								■	■					
Implementation – Core features (roles, CRUD)									■	■	■			
Implementation – Interactivity (JS, responsive)										■	■	■		
Testing & debugging											■	■		
Final report preparation & submission												■	■	
Project presentation & demo														■

4. Restrictions and Academic Integrity

- The use of web page creation tools (e.g., Wix, WordPress, Weebly, Squarespace, etc.) and pre-built templates or frameworks (e.g., Bootstrap, Foundation, TemplateMonster, ThemeForest, etc.) is strictly prohibited.
- All work must be your team's original work. Copying code, designs, or content from other students, online repositories, or external sources without proper acknowledgment will be treated as plagiarism.
- AI-assisted tools (e.g., ChatGPT, GitHub Copilot) may be used only for idea generation or debugging guidance. However:
 - All final code, designs, and documentation must be written and explained by your team.
 - You must be able to demonstrate and explain your code during the presentation.
 - Any code or text generated with external assistance must be acknowledged in your report under "References / Tools Used."

5. Submission

All deliverables must be submitted to Moodle in the following format:

- **Documents** – Microsoft Word/ PDF format.
- **Web application** – Compressed ZIP file containing all source code, database export (SQL file), and assets (images, icons, fonts, stylesheets (CSS), scripts (JavaScript) etc.)

6. Assessment Criteria

Documentation (50%)

Criteria	Max Marks	Not Available	Needs Improvement	Satisfactory	Good	Excellent
Introduction	5	0 – No introduction provided.	1–2 – Vague or unclear; missing project purpose/scope.	3 – Basic intro; purpose/scope mentioned but lacks detail.	4 – Clear intro; purpose/scope explained with some detail.	5 – Highly clear, concise intro; purpose/scope thoroughly explained and engaging.
Project Plan & Requirement Gathering	5	0 – No plan/requirements.	1–2 – Weak plan; missing objectives, users, or Gantt; vague requirements.	3 – Basic plan with objectives and some requirements; minimal justification.	4 – Clear plan with most elements; requirements documented with some justification.	5 – Comprehensive plan with objectives, scope, users, Gantt; well-justified requirements aligned with goals.
Storyboard, Modeling & Database Design	15	0 – No storyboard, models, or database design.	4–6 – Incomplete or unclear storyboard/ERD/wir eframe; weak data dictionary.	7–9 – Basic storyboard and ERD provided; some database details included.	10–12 – Clear and accurate models (wireframes, flowcharts, ERD, dictionary); mostly complete.	13–15 – Highly accurate, professional models; well-documented DB (normalized ERD, complete dictionary, smooth integration).
Code Snippets Explanation	15	0 – No code snippets or explanation.	4–6 – Few or unclear snippets; explanations missing or superficial.	7–9 – Basic snippets included; explanations partially clear.	10–12 – Good variety of snippets; clear explanations showing functionality.	13–15 – Comprehensive, well-chosen snippets; excellent explanations; strong

						coding understanding.
User Manual	5	0 – No user manual.	1–2 – Poorly written; unclear instructions.	3 – Basic manual included; covers some features.	4 – Clear manual covering most features with usable instructions.	5 – Comprehensive, professional manual; step-by-step guidance for all roles.
Conclusion, References & Documentation Standard	5	0 – No conclusion/reference s/formatting.	1–2 – Weak conclusion; poor references; poor formatting.	3 – Basic conclusion; some references; acceptable formatting.	4 – Clear conclusion; relevant references; good formatting.	5 – Strong conclusion; high-quality APA references; professional formatting and structure.

Implementation (50%)

Criteria	Max Marks	Not Available	Needs Improvement	Satisfactory	Good	Excellent
Web Page Presentation, Navigation, Usability & Responsiveness	5	0 – No design/navigation.	1–2 – Poor layout/navigation; usability issues; not responsive.	3 – Basic layout/navigation; partially responsive.	4 – Clear design; good usability; mostly responsive.	5 – Professional design; intuitive navigation; excellent usability; fully responsive.
HTML & CSS	5	0 – No HTML/CSS used.	1–2 – Poorly structured HTML/CSS; invalid tags; inconsistent styling.	3 – Basic HTML/CSS with some errors; limited responsiveness.	4 – Good structure and styling; responsive with minor issues.	5 – Excellent, semantic HTML; professional CSS; fully responsive; accessibility best practices.

Features using JavaScript	5	0 – No JS features.	1–2 – Poor/broken JS features; minimal interactivity.	3 – Basic JS features; limited interactivity.	4 – Functional, useful JS features; responsive performance.	5 – Advanced, well-integrated JS features; excellent performance, smooth interactivity.
Features using PHP & Database Integration	25	0 – No PHP or DB integration.	8–12 – Minimal PHP; weak DB connection; poor validation.	13–16 – Basic PHP; CRUD partly working; DB partially integrated.	17–21 – Good PHP implementation; CRUD functional; secure DB integration with minor issues.	22–25 – Robust PHP; full CRUD; secure DB integration with validation, error handling, optimized queries.
Presentation	10	0 – No presentation.	2–4 – Poor presentation; unclear delivery; lack of teamwork.	5–6 – Basic presentation; limited clarity; partial teamwork.	7–8 – Clear presentation; teamwork evident; good demo with minor issues.	9–10 – Highly professional; engaging delivery; strong teamwork; smooth demo with confident Q&A.